

ENVIRONMENT

TEMPLATE

ENVIRONMENT	
Importance wala question highly productive - Supporting Biodiversity, shelter to various species - carbon sink - Regulating services like Purify, micro climate, soil/coastal erosion, biogeochemical cycles & nutrient recycling, diseases - Provisioning: Fuelwood, local economy - Fisheries, eco tourism - cultural - aesthetic, recreational, spiritual, tribal - Natural barriers - Bioindicator	
Challenges Threats: - natural Cyclone Forest fire Floods El - Nino / Land use change - manmade - 1. Deforestation - Land use change / encroachment 2. Habitat destruction from MATTIE - Mining, agri, industry, transport, industry ; HEP 3. Unsustainable tourism 4. Waste dumping / pollution - agri run off ; industry ; sewage ; municipal solid waste ; plastic 5. Unsustainable urbanisation : population explosion, consumerism, concretisation, linear economy 6. tragedy of commons / over harvesting 7. Climate change 8. Invasive species	
Impact - hydrosphere • surface water - eutrophication ; dead zones ; contamination • ground water : depletion • open oceans : sea Level rise ; acidification ; de oxygenation ; temp rise ; stratification ; pollution ; increased cyclones ; AMOC • coastal : receding / emerging coast ; salt water intrusion - atmosphere • extreme weather events • global warming • heat islands • acid rain • smog • air pollution / AQI	- biosphere • 6th mass extinction • man animal conflict • changing migration patterns of animals • pest attacks • bio magnification and bio accumulation • co extinction - lithosphere • desertification • erosion • salinity • soil moisture desertification - anthroposphere • food security • loss of livelihood • disaster - GLOF • pandemics / diseases • permafrost • refugee crisis • social - PHEE WASH • agri distress

	• loss of productivity • human health • water security • accidents of life
Solutions • One health • Practices - - Bottom up - Govt - ESIA ; 48 A ; 21 - individual -5R(refuse ; reduce ; reuse ; recycle ; repair), LiFE ; 51A Business - SDG 12 triple bottom line (people ; profit ; planet), Pay for ecosystem services model ESG Principles - - Policy: Wise use, Precautionary, Ecocentric - Justice : CBDR RC, L&D, absolute liab, polluter pays - Corporate: triple bottom line - people profit planet, ESIA, Access Benefit sharing (fair earth share) Collective action vs collective suicide	Principles - Wise use, Precautionary, Ecocentric, Access & Benefit sharing (fair earth share), CBDR-RC, (for justice) Loss & Damage, Pro planet people, absolute liability Polluter pays principle Climate adaptive, climate responsive & climate resilient development LT Low Carbon Development strategy (COP 27) - multi sectoral measures for net 0 & climate resilience SaptRishi - Green Growth Climate crisis to climate emergency Think Global act local Lab to street

1. COMMON INTROs, CONCLUSIONS:

- ___ is not just an option but a necessity for a sustainable future
- Collective action vs collective suicide - Antonio
- Saptrishi - Green Growth

2. SUSTAINABLE DEVELOPMENT

- Sustainable development represents a harmonious relationship between , development , social equity and environment
- Sustainable development report : India 121 - SDG
- 168 - environment performance index
- Triple emergency - CC ; biodiversity loss and pollution
- India voted in favor @ UNGA to recognize right to clean , healthy and sustainable env as human right
- "The greatest threat to our planet is the belief that someone else will save it." - Robert Swan
- "We won't have a society if we destroy the environment." - Margaret Mead

3. BIODIVERSITY

a. Data

- i. IUCN : 1/3rd of species are threatened with extinction
- ii. India a megadiverse country
 - 1. 2.4% of land - 7-8% of all recorded species
 - 2. 4/36 biodiversity hotspots
- iii. Living planet report : 70% wildlife population declined since 1970

b. Triple planetary crisis : CC ; pollution (DALY) ; biodiversity loss (6th mass extinction)

c. Amur Falcon Model - Predators to Protectors - awareness generation to reduce poaching

d. Millet Sisters network to protect indigenous crop diversity

e. Evil quartet

- i. Habitat loss and fragmentation : CC ; desertification ; developmental project
- ii. Over exploitation - extinction(stellar's sea cow) ; illegal trade
- iii. Invasive species - eg lantana ; 10 invasive species in India - damage > rs 8 trillion
- iv. Co-extinction

f. ESZ

- i. ESZ are ecologically important and fragile areas around protected areas designated to be protected from industrial pollution and unregulated development under EPA 1986
- ii. Mark Dowie : the only thing that displaced more people than was is wildlife conservation
- iii. WF

1. Periyar conservation model

- a. Eco tourism model
- b. Eco development committee
- c. Integrated local tribals into protection and conservtaion

4. LAND

a. STATS

- i. India supports 16% of world population and 20% livestock population on 2.5% of its geographical

area

- ii. Inefficient land use - agri : bulk of land use but only 16% contribution to GDP
- iii. Avg land holding size is 1.08 ha
- iv. Property dispute : NITI - avg 20 yrs to settle a dispute

b. T Haque committee - model land leasing act

c. DESERTIFICATION

- i. deterioration of the quality and productivity of land due to various factors including climatic variation and human activity
- ii. UNCCD ; 30by30 target ; bonn challenge
- iii. Campa ; green india mission ; urban forestry : Pune Wajre forest model ; iForest model
- iv. Great green wall - Sahel region
- v. Data
 - 1. 30% of total land area threatened with degradation in india - ISRO
 - 2. **MoEFCC** - 2.5% of GDP already lost due to desertification
 - 3. **Since 1850s** - land under cultivation has become 3x
- vi. BP : **Project BOLD/Bamboonomics** - Bamboo oasis on lands in drought in arid and semi arid regions - Tribal livelihood + erosion control

5. ECOSYSTEM

a. Remedy for all

- 1. Paris ; landscape approach ; secure himalayas
- 2. Humans : ESZ ; eco tourism ; community participation ; kasturiranjana committee ; sustainable agriculture
- 3. Technology - drones ; satellite ; EWS

b. FOREST

i. Data

- 1. FSI - forest cover - 21.7% (target NFP 1988 - 33%)
- 2. 3 lakh ha forest land diverted in 15 yrs - max for mining ; road or irrigation
- 3. Antonio guetres : we have a choice
collective action of collective suicide

4. 45% of earth's original forest lost in last century

5. UNFCCC : 80% deforestation in tropical countries due to agriculture

ii. BP

1. Eco feminism: Piplanatri

2. JFM - Maldharis

c. MANGROVES

i. Salt tolerant vegetation that grows in intertidal estuaries

ii. Mangroves for future ; mangroves atlas

iii. Data

1. ISFR report 2019 : Mangrove cover in Indian sub continent : ~ 5000 sq km (increased 1.1%)

2. Map : sunderbans ; AN ; cauvery ; KG basin ; mahanadi basin ; ratnagiri ; GA ; kori creek ; kutchh

iv. SAMIE initiative - SUSTAINABLE AQUACULTURE IN MANGROVE

ECOSYSTEM - Planting mangroves around shrimp ponds instead of clearing them - no need for fish feed + active mangrove conservation

v.

d. WETLAND

i. Defn : RAMSAR - areas of submerged lands such as marshes , swamps , bogs and fens usually upto 6m water depth

ii. 7 lakh wetlands in India

iii. In the **last 100 years - 65% disappeared**

iv. **Deepor Beel** - 1/3rd encroached

v. Principles : wise use

vi. BP : Keshopur Maini community reserve

vii. WF : need for scientific criteria for identify wetlands + data bank + cooperative and decentralised efforts

e. COASTAL

i. The indian coastline runs over a distance of 7500 km and has more than 1200 islands ; 1/5th of the population and 3 metropolitans

ii. Coastal erosion

1. Wearing away of land and removal of beach sediments by high winds ; drainage , wave action , wave currents and tidal currents
 2. MoEFCC : 40% of Indian coastline is subjected to high , medium or low coastal erosion
 3. "Kerala Coastal Zone Management Project" : integrated coastal zone management practices to address erosion, promote sustainable development, and conserve coastal ecosystems.
 4. Geo textiles
- iii. Climate action plan for coastal areas
1. A climate action plan is detailed and strategic framework for measuring , planning and reducing GHG emissions and related climatic impacts and provides preventive measures to address the negative outcomes of climate change
 2. BMC is drafting a mumbai climate action plan (MCAP)
 3. Steps taken
 - a. NAPCC
 - b. CRZ
 - c. National coastal mission
 4. Impact of climate change on coastal areas has strong strategic , economic and political repercussions

f. CORALS

- i. Coral reefs are large underwater structures of calcium carbonate made of skeletons of corals
- ii. Map : Malvan , Mannar ; kutch ; AN ; LD
- iii. Tropical rainforest of the oceans
- iv. Data
 1. India : home to 30% corals ; 50cr ppl depend for livelihood and cultural identity
 2. 6th status of the corals of the world report - 14% of corals reef lost since 2010
- v. ICMAM (integrated coastal and marine area mgmt) ; INCOIS : coral bleaching alert system ; BioRock ; Reefwatch

vi. BP : cryomesh

g. MOUNTAIN

h. WESTERN GHATS

i. UNESCO world heritage site which spans along india's western coast encompassing rich biodiversity and unique ecosystems

ii. Difference in estimation : Gadgil committee : 100% ESA ; kasturirangan panel : 37% as ESA

i. CRYOSPHERE

i. Frozen component of the earth system that are at or below the land and ocean surface eg : permafrost

ii. Permafrost holds 2 times as much carbon as atmosphere

iii. 10% of the world population lives in high mountains

iv. Glaciers - Thwaites, Larsen C, Ojukull (Iceland) - 80% expected to be lost by 2100 as per IPCC

v.

j. OCEANS

i. Ocean acidification

1. Decrease in pH levels in oceans <8 due to CO₂ uptake

ii. AMOC

1. It is the atlantic branch of global ocean conveyor belt which distributes heat and nutrients throughout the world

2. IPCC : AMOC is very likely to decline over 21st CE

6. ANTARCTICA

a. Antarctica treaty 1959

i. x military activity ; x nukes ; x territorial claims

ii. Scientific research allowed

b. world 's 90% ice ; 70% freshwater

c. Indian antarctica Bill ; 2021

d. Dakshin Gangotri ; maitri ; bharti

7. POLLUTION

a. AIR POLLUTION

i. 5th most polluted country

ii. WHO : 14/15 most polluted cities are in India

iii. CSE : life expectancy in India decreased by 2.5

yr due to air pollution

iv. As of 2020 - 90% TPPs not compliant with sulphur standards, > 51% not compliant with PM standards

v. Steps

1. NCAP ; CAQM ; EPCA ; smog towers
2. Fly ash mgmt rules 2016 (India - only 63% fly ash utilisation)

vi. Best practice : london ; beijing

vii. Transport sector

1. 23% CO₂ ; 60% by 2050
2. 3rd most CO₂ emitting sector
3. 90% emission for road
4. Vehicles expected to double by 2030 (200 million)
5. FAME ; BS VI

viii. Agri

1. Agri + livestock : 18% of GHG

ix. Cities contribute to 60% of GHGs & 70% of GDP

x.

8. WATER

a. Data : india 4% freshwater ; 17% population ; dd will outstrip ss by 2030 : NITI

i. Per capita availability : 1951 - 5177 ; 2011- 1545 ; 2030 - 1300

ii. 70% contaminated with heavy metals

b. Pak - Increased domestic violence instances increased due to water insecurity & heat led aggression

c. Sahel - conflicts due to reducing fertility from water stress

d. Water wives

e. Govt steps

- i. Jal shakti mission
- ii. MGNREGA : recharging aquifers
- iii. Atal bhujal
- iv. River interlinking
- v. Water act

f. GROUND WATER

i. WB : India's groundwater depleting at 0.4 m / yr

ii. 80 % of rural domestic water needs ; 50% urban

water needs

iii. 90% used for agri

iv. Thomas Fueller - we never know the worth of water until the well is dry

v. Govt efforts

1. CGWB / CGWA

g. Jal jevan mission

i. Budget allocation increased by 23%

ii. WHO : 4 lakh diarrhoea deaths can be averted if JJM meets its target

h. WATERSHED DEVELOPMENT

i. Watershed development means conservation and regeneration and judicious use of all natural resources in in the watershed

ii. Scheme

1. MGNREGA ; neeranchal ; PMKSY ; jal shakti abhiyan

iii. BP : Hiware Bazar ; ralegaon siddhi ; Jal Swavalamban (RJ), Jalgaon model of Pani Panchayat

iv. Integrated watershed management - NGOs Pani Foundation, Sujala (KN)

v. Micro-watershed atlas of India 2019

vi.

i. Eutrophication

i. Excessive plant and algal growth due to increased availability of one or more limiting growth factors needed for photosynthesis such sunlight CO₂ and nutrients

j. WF

i. New agronomic practices : agroforestry , ZBNF and sub surface irrigation

ii. Mihir shah committee

1. Dd mgmt - crop diversification ; RRR

2. Ss side mgmt : nature based solutions ; dam mgmt ; river mgmt

3. Water governance : multi stakeholder national water commission

9. SOLID WASTE MANAGEMENT

a. Data

i. Waste generated : 65 MT / yr ; 165 MT by 2030

ii. Out of 63 MT ; 43 collected ; 13 treated

iii. Waste to wealth

b. WASTE WATER

i. BP : kolkata sewage fed aquaculture ; Avadi sewage plant TN

c. E WASTE

i. Defn : all items of electric and electronic equipments(EEE) and their parts discarded after use

ii. India 3rd largest generator (3.2MT); 17% collected and recycled

iii. Best practice

1. LIAM apple : recycling robot
2. EU : Right to repair
3. Bhopal eWaste clinic
4. EPR

iv. Basel convention : control of transboundary movement - includes e-waste

d. MARINE LITTER

i. 8 million tonnes of plastic end up in ocean every year

ii. Plastic is 80% of all marine debris

iii. Bioaccumulation and biomagnification : microbeads in the blood of new born

iv. Cocos keeling island

v. Indian efforts

1. Plastic waste management rules 2018
2. KL's suchitwa mission - fisherman finding and collecting plastic
3. Puneet sagar abhiyan

vi. International

1. GloLitter partnership project : towards low plastic future
2. Ocean clean up
3. MARPOL by IMO
4. SDG 14

e. PLASTIC POLLUTION

i. Data

1. India - 9.3 MT / yr - 43% SUP
2. Only 30% recycled
3. Global plastic outlook (OECD) - plastic waste 3x by 2060

ii. Steps

1. Plastic waste mgmt amendment rules 2021
2. India plastic challenge - hackathon 2021
3. Puneetsagar abhiyan
4. EPR on plastic packaging

iii. BP

1. AP : rice for plastic ; Ireland : Plas Tax
2. Precycling practice ; south korea - bringing own reusables to supermarkets ; cafe
3. Main indoori jholadhari
4. MP Bhartan bhandar
5. Wayanad - using plastic in road construction

iv. China and malaysia have banned illegal plastic imports from USA , canada , UK

f. CIRCULAR ECONOMY

i. Defn : it is an economy where products are designed for durability , reuse and recyclability

ii. 6 R

1. Reduce
2. Reuse
3. Recycle
4. Refurbish
5. Recover
6. Repair

iii. Steps :

1. Vehicle scrappage policy ; right to repair bill
2. National productive council push for circular economy
3. India - Australia circular economy hackathon (I-ACE)
4. Digital india prog - significant component of recycling of e wastes

iv. International

1. China - circular economy promotion law
2. Germany and japan - binding principle for reorganization of economy
3. SDG 12

v. Conclusion : take , make , dispose mindset to 6R

g. BIO MEDICAL WASTE

i. Data

1. Drastic increase during covid - 20T/ day to

200T/day - CPCB

2. In general only 15% BMW is hazardous

ii. Defn

1. Human and animal anatomical waste ;
treatment apparatus like needle ; syringes etc
2. Colour coded categorisation - blue ; red ;
yellow ; white

iii. BMW rules 2016

10. LAND SUBSIDENCE

- a. Defn : it is shrinking of the ground because of
underground material movement ; it can be caused by
gradual setting or sudden sinking of the earth's surface
- b. Committee : Mishra committee 1976
- c. BP : vetivar system for soil erosion

11. LAWS

a. FRA

i. 2023 amendment

1. Defines forest in objective manner
2. No prior approval for strategic , security or
defence projects in 100kms of international
border

ii. The act in its current form has created several
hurdles in the way of development ; yet any
change will only work if it recognises the
symbiotic relationship between forest ,
biodiversity and the stakeholders

b. NGT

i. PoNJ

ii. Ennore oil spill : which affected >1 lakh
fisherman ; NGT order the shipping company to
pay 240cr compensation to affected people

iii. Green hurdle ; selective adjudication

c. PPV&FR ACT

i. PepsiCo case

d. WPA

i. Introduction: india is cradle of wildlife from the
times of IVC ; it has been worshipped as gods
and goddesses ; WPA 1972 was enacted to
preserve and promote the wildlife of India

ii. Proposed amendments in the act

e. EIA

- i. Defn : process or study which predicts impact of proposed project on envt ; prevents approval w/o considering impact and compares alternatives to decide best one to minimise impact
- ii. Flow diagram : screening → scoping → impact analysis → mitigation → draft EIA report → public hearing → revision & approval → implementation → monitoring
- iii. Data
 1. 41%NGT cases about faulty EIA
- iv. -ves examples
 1. LG polymers : no EIA for decades
 2. Vadavan port EIA - considered only 1 season's data
- v. Kasturiranjana committee : GRM for violation in compliance ; capacity building of stakeholder ; no activity in ESZ
- vi. WFA : SEA : strategic environment assessment

f. BDA

- i. Was enacted to ensure conservation , benefit sharing and sustainable use of bio diversity in India in line with CBD and subsequent protocols like Cartagena for biosafety and Nagoya for ABS
- ii. Issues
 1. Commerce over conservation
 2. Too much exception
 3. ABS - Divya pharmacy case
- iii. Conclusion : biodiversity is the reason for survival of humankind and the world must collaborate to make targets of global biodiversity framework into reality

12. ENERGY

a. Data

- i. 3rd largest energy consumer
- ii. Highest growth rate of energy consumption - 11%
- iii. Since 1850s - energy use has become 25x
- iv.

b. Energy trilemma - energy equity ; energy security ; sustainable energy

c. SOLAR

- d. ISA
- e. NATURAL GAS
 - i. Target 15%
 - ii. Cleanest among fossil fuels
 - iii. 50% is imported
 - iv. TAPI pipeline
- f. GEOTHERMAL
 - i. 1st ever geothermal field development - Puga Valley , LA
 - ii. Geothermal energy is the energy generated and stirred inside Earth's crust
 - iii. Data
 - 1. 350 sites - potential of 10GW
- g. METHANOL ECONOMY
 - i. Methanol economy is based on the concept of replacing dependence on oil and coal with methanol
 - ii. Slightly lower energy content than petrol and diesel
 - iii. NITI roadmap : substitute 10% of crude imports by 2030 by methanol alone
- h. BIO-ETHANOL
 - i. E20 by 2025
 - ii. PM JIVAN
 - iii. Govt should work on an incentive scheme to promote manufacture and use of "flex-fuel" engines in vehicles
- i. PM KUSUM
 - i. Target to add 30.8 GW solar capacity + de-diselisation + feeder solarisation

13. CLIMATE CHANGE

- a. Climate change
 - i. Defn : significant changes in global temperature , precipitation , wind pattern and other measures of climate that occur over several decades
 - ii. Panchamrit
 - 1. Net zero by 2070
 - 2. Non fossil energy - 500 GW by 2030
 - 3. 50% renewable energy by 2030
 - 4. Reduce total carbon emission by 1 bn tonnes by 2030

5. Reduce carbon intensity by 45% by 2030
- iii. Great Thunberg : “ if solutions within the system are so impossible to find ; we should change the system itself “
 - iv. Antonio Guterres : IPCC reports are code red for humanity
 - v. Climate paradox : china - highest emission but part of G77
 - vi. Energy trilemma : energy equity ; energy security ; environmental sustainability
 - vii. Keywords : 6th mass extinction ; DALY ; degrowth ; LiFE ; circular economy ; carbon colonialism ; climate denialism ; climate proofing ; greenwashing ; hothouse ; tipping point
 - viii. Data
 - 1. Atlas of energy : annual CO2 emission from fossil fuels
 - a. China - 10 gigatonnes
 - b. USA - 5 gigatonnes
 - c. India : 2.3 gigatone
 - 2. Per capita emission
 - a. US : 14 MT
 - b. China 7 MT
 - c. India 1.78 MT
 - 3. India ranked 8th - climate change performance indicator - german watch
 - 4. India 7th most vulnerable country in Climate Risk Index (CRI) - #1 in fatalities and 2nd highest monetary losses
 - 5. 1 person of USA = 108 people of ethiopia : carbon emissions
 - 6. Carbon -ve countries : bhutan
 - 7. IPCC - to achieve 1.5 degree target ; rapid reduction of GHG emission 43% by 2030 relative to 2019 is required
 - 8. MoES Assessment Report:
 - a. heatwaves to become 3-4x by 2100, marine heatwaves to become 2x (IPCC)
 - b. drought frequency to double in next decade
 - c. sea level to rise by 3.3 mm / year

9. Health care cost of climate change - 8.5% of GDP

10. Productivity loss as per ILO - 34 million full time jobs

11. Sea level rise - 20% of the global assets exposed

ix. Best practice

1. Meenangadi village KL

a. Efforts for carbon neutral panchayat

b. GHG inventory

c. Tree bank project - loans as per trees planted (no collateral); panchayat pays the interest

2. Adaptation - NEED : North European Enclosure Dam

b. Climate justice

i. Pak floods - Cost > 30 Billion \$, 1700 casualties

ii. 58 Climate vulnerable countries ----> 5% global emmissions

iii.

c. COP 27

i. Sharm el-sheikh implementation plan

1. Breakthrough in L&D

2. Proposed EWS to all population

3. FAST - food and agriculture for sustainable transformation

4. Koronivia Joint work for agriculture

5. ENACT initiative : enhancing nature based solutions for accelerated climate transformation

6. Mangrove breakthrough - protect 15mn ha by 2030

7. African COP

ii. Contention

1. No commitments to phase out all fossil fuels

2. Inadequate climate finance

3. Lack of clarity in L&D

4. Greenwashing

5. Fossil fuel lobby

iii. Indian concerns

1. Gap in operationalisation of equity and

- climate justice
- 2. Rapidly shrinking global carbon budget
- 3. Due consideration to developmental needs of developing countries
- 4. Unilateral measure and discriminatory practices - CBAM
- 5. Extend scope of mitigation to agriculture
- 6. Selective singling out of source w/o any basis in the available science
- 7. Shortage in climate finance

d. CLIMATE FINANCE

i. Carbon finance is local , national or transnational funding from public , pvt and alternative sources that seeks to support climate change mitigation and adaptation actions

ii. Key words : carbon leakage ; coal cess ; CBAM by EU ; CBDR ; zero sum game

iii. Data

- 1. \$ 70-100 bn / yr needed by developing countries for adaptation b/w 2010-2050
- 2. Copenhagen accord : \$100 bn p.a. By 2020
- 3. <2% of promised funds given in last 10 yrs
- 4. India needs \$10 trillion investment to achieve net zero by 2070
- 5. Opportunities
 - a. \$1 investment = \$4 return
 - b. Green jobs - 65mn new jobs can be created
- 6. 3/4th adaptation projects have no direct co-relation

iv. Efforts

- 1. Int : GEF ; GCF ; adaptation fund
- 2. Catastrophe bonds

v. RBI Dy. Governor:

- 1. Setup **GIIT - Green Infra Invt Trust** on lines of REIT, INVIT
- 2. Taxonomy on **green finance** to avoid **greenwashing**
- 3. Expertise to evaluate climate risks in **green loan portfolios**

e. NET ZERO

- i. Refers to a balance where emissions of GHGs are offset by the absorption of an equal amount from the atmosphere
- ii. India - 2070
- iii. New NDCs
 1. Reduce emission intensity by atleast 45%
 2. 50% non fossil fuel energy
 3. 2.5-3 bn tonnes of additional carbon sink
- iv. Steps
 1. OSOWOG
 2. IRIS - infrastructure for resilient island states
 3. CDRI

f. LT-LEDS

- i. Lays out the steps to achieve net zero by 2070
 1. Low carbon electricity system : 50% non fossil ; RPOs ; 3x rise in nuke installed capacity by 2032
 2. Transport : 20% ethanol blending ; railways net zero by 2030
 3. Urban efficiency : national urban policy framework framework
 4. Innovation + low emission industry : green H2 ; national mission for enhanced energy efficiency
 5. CO2 removal : CCUS
 6. Forest cover : NDC target : 2.5-3 bn tonnes of CO2 equivalent by 2030 ; green indian mission
 7. Economic and financial aspect of low carbon development : business and

g. MISSION LiFE

- i. Objective : to promote environmentally conscious lifestyle focused on ‘mindful & deliberate ‘ utilisation instead of ‘wasteful & mindless consumption ‘
- ii. India led global mass movement which aims to nudge individual and collective action to protect and preserve environment
- iii. >80% of all RLB and ULB to become

environment friendly by 2028

iv. Change in dd $\leftarrow \rightarrow$ change in ss $\leftarrow \rightarrow$ change in policy

v. PM Modi :

1. LiFE does not look at which country is in coalition but brings everyone together
2. Mission LiFE is an important step towards renewed and reimagined multilateralism

vi. BP : solar powered village of Modhera

h. INDIA'S CLIMATE PERFORMANCE

i. AGRICULTURE AND CLIMATE CHANGE

i. Data

1. 18% GHG emissions from agriculture
2. 70% agri energy usage in irrigation
3. ES 2018 : CC will cause loss of farmer's income b/w 15-18% in irrigated areas and 20-25% in unirrigated areas
4. GM crops
 - a. Only 29/170 countries grow GM - even after 25 years
 - b. 90% in 5 countries - USA, Canada, Brazil, Argentina India
 - c. 5th largest area under GM cultivation - 11 mn ha

ii. ORGANIC FARMING

1. Organic farming is a sustainable agricultural approach that avoids synthetic inputs and emphasises soil health, biodiversity, and natural processes to produce food.
2. Data
 - a. India ranks 8th in terms of World's organic agricultural land and 1st in terms of total number of producers as per 2020 data
 - b. Export of organic food products rose by 51% in terms of value
 - c. Only 2% of the 140 mn ha net sown areas
3. Govt steps
 - a. PGS - India certification
 - b. Paramparagat krishi vikas yojana

c. NMSA

4. Awareness campaign similar to SBM for “swachh food “

iii. SUSTAINABLE FOOD SYSTEMS

1. Defn : SFS is a food system that delivers food security and nutrition for all in such a way that it is
 - a. Economically sustainable - profitable
 - b. Socially sustainable - broad based benefits for society
 - c. Environmentally sustainable - +ve or neutral impact
2. Efforts
 - a. NMSA ; per drop more crop ;
paramparagat krishi vikas yojana
3. Conclusion : precision agri ; agro climatic agri ; adoption of tech

j. CARBON TRADING MARKET

- i. It is a market-based approach to reducing GHG emissions, particularly CO₂; that contributes to climate change. Its main goal is to provide economic incentives for industries and entities to limit their emissions of these gasses.
- ii. Issues with carbon markets :
 1. Double counting of emission reduction
 2. Low market transparency
 3. Greenwashing
- iii. India
 1. REC
 2. Energy saving certificates - under PAT
- iv. Conclusion : gradually move from PAT to cap and trade

k. IPCC REPORT

- i. Avg annual emission 2010-19 higher than any previous decade ; but rate of growth slowed down
- ii. Global share of emission from urban areas is increasing
- iii. Suggestions
 1. Mitigation efforts to be embedded within wider development context
 2. Equitable partnerships - civil society ,

- political actors ; buisnesses ; youth
3. Avoid Maladaptation : by inclusive planning
; considering biodiversity and autonomous adaptation
 4. Climate resilient development
- iv. Impact on food production
1. 69% indian marine fisheries impacted by CC
 2. Rice production can decrease by 10-30 %
 3. Maize production can decrease by 25-70%
- v. 2019 - india : 4mn disaster induced displacement
- vi. Conclusion : a brief and rapidly closing window of opportunity to secure a liveable and sustainable future for all

SECURITY - INTEGRATE FROM ANSWERS

TOPIC	Intro	Conclusion	Value Addition
General IS challenges	India is highly vulnerable to security threats due to its location at the epicenter of terrorism & instability, as a drug transit nation with a long vulnerable coastline and porous and contested borders.	Chanakya - Need 3 S for holistic national power - Shastra - military prowess Shaastra - academia (Book to Battlefield & Civil-military fusion - Ex: China) Sanskriti - ideals and principles <u>Multi-domain operations / Whole of Govt & society approach</u> The fight against ___ is not that of any one govt/ agency but requires all nations, & stakeholders to come on same platform & make coordinated efforts with equal intensity & seriousness. If implemented well _____ step should go a long way in tackling the issue of _____.	<u>Issues & Solutions:</u> - S - security forces - cap building, fencing - E - Econ/Finance - C - Coop - bilateral/intl - U - upgraded surveillance - R - Regional development/ Good govnc - I - Intelligence (locals), Infra/ involve PPP - T - trade/tech - Y- youth/ WHAM
Border security	India has a land border of > 15000 km with 7 countries of which _____ poses unique challenges for internal security.	Need to view border security from a holistic prism of vibrant P2P exchanges, upgraded surveillance using cutting edge technologies & robust	Robert Frost - Good fences make good neighbours

		<u>bilateral cooperation to reduce differences.</u>	
<u>China border</u>	India shares 3488 Km of border with China which remains disputed at multiple locations <u>Eg: Recent Ladakh standoff</u>	<u>Vibrant village program</u>	Cartographic & Nomenclatural aggression
<u>Nepal Border</u>	Open border with Nepal - with no travel restrictions to citizens on either side		
<u>Bhutan Border</u>	<u>Same as Nepal</u>		
<u>Myanmar Border</u>	The India- Myanmar border runs along 4 states in NE & is marked by Free Movement regime upto 16 km on either side due to deep ethnic cultural ties.		
<u>Pak border</u>	Pakistan is India's northwestern neighbor sharing maritime & land boundary demarcated based on the Radcliffe award of 1947.	Madhukar Gupta committee	- Village defence guards - CIBMS
<u>Bangladesh Border</u>	India shares a long porous land, riverine & maritime border with Bangladesh along India's east and north eastern region.		
<u>Overall Defence</u>	India's defence budget is the 5th largest in the world - due to threats to sovereignty and integrity from its hostile neighbours.	General Manoj Narawane - Need <u>National security strategy,</u> <u>national defense strategy,</u> <u>higher defense org in place</u>	<u>Keywords:</u> - Critical deficiencies - Force multiplier - <u>Combat edge</u> Defence exports - 1.5 Billion \$ vs global market - 500 Billion \$
<u>Defense indigenisation</u>	Process of developing and producing essential military capabilities within a nation's own borders <u>SIPRI - India is the world's largest arms importer</u>	With a looming threat of two front war, ensuring Atmanirbharta in defence capabilities is an urgent necessity. With 60% dependence on imports, we need to balance defense needs of	Exports: > 6x increase in last 5 yrs > 75 countries <u>Issues:</u> <u>Delay</u> - INS Vikrant - 1st indig aircraft carrier 2009-2021 <u>China</u> - supercarriers

	60% weapons foreign origin - CAG	<u>urgency vs autonomy.</u> <u>Steps:</u> - Defence corridors <u>Brahmos in Lucknow</u> - <u>AK 203 MII</u> - DPEPP 2020 - Defence prod & export promotion policy - <u>DAC</u> - <u>FDI 74% automatic</u> <u>Solutions:</u> - Dhirendra Singh committee - Prioritise <u>Buy (Indian IDDM)</u> - Defence Bureaucracy - Long term perspective plans	2x number & 2x capacity <u>Policy uncertainty</u> - Ex: DPP changed 7 times since 2001 <u>Risk</u> - Rel Defence bankruptcy - lost to Mazgaon <u>DRDO</u> - no specialization <u>Acquisition trilemma</u> - time, cost, quality <u>Low servability rates</u> - (CAG - Sukhoi - 30- 35%) <u>Pvt - only 20% share</u> vs DPSUs 80% <u>Capex spending - only</u> 25% of defense budget
<u>Maritime security</u>	India has a long coastline of 7517 kms and > 1200 islands making it particularly vulnerable to maritime security threats	India's vision of "Neel Kranti se Arth Kranti" & rising threats from China in the IOR call for building domestic capacity along with International push for Russia's proposal of a dedicated maritime security body in UN Security Council.	
<u>Cyber security</u>	Securing digital systems, networks, and data from cyber threats to ensure confidentiality & integrity.	I dream of a digital India where cyber security becomes an integral part of national security - PM Modi	Internet security threat report - India 3rd most vulnerable to cyber security threats 15% increase in cyber crimes in 2021 over 2019 (NCRB)
<u>Radicalisation</u>	Process by which an individual comes to adopt extreme political or religious ideas & aspirations	<u>Aquaba process of Jordan</u> Maharashtra Model of <u>3 pronged counselling</u> - family,clergy,professionals + reintegration	
<u>New forms of militancy</u>	<u>Over Ground workers</u> - help militants in movement, ammunition supply & accommodation <u>Hybrid militants</u> - not listed as ultras but radicalised enough to carry out terror strike then slip back to routine life Ex: sleeper	SL restorative justice Model of vocational, spiritual, social & psychological rehabilitation of LTTE militants into the mainstream.	Keywords: Faceless militancy <u>Mission Pehal</u>

	<u>cell</u> <u>Lone wolf</u>		
<u>Money Laundering</u>	Process by which illegally obtained money is made to appear as if originated from legitimate source	<u>Vijay Mohanlal Choudhary vs UOI</u> - ML is a heinous crime which only affects socio-economic fabric of a nation but promotes other heinous crimes	IMF - 2-5% of world GDP is landered <u>Placement - layering - integration</u> Steps: - <u>PMLA Rules 2023</u> - stringent KYC norms - <u>Benami transactions prohibition Act</u> - <u>PMLA Act 2003</u> - cognisable & non-bailable + confiscation of proceeds of crime - <u>Black Money Act</u> - Deregistered <u>shell companies</u>
<u>Fake news</u>	Info that has been deliberately fabricated & aimed at deceiving others into believing falsehood	<u>KL model of school education</u>	
<u>LWE</u>	LWE / Naxalism denotes radical ideologies advocating for socio-political change through armed struggle against established authorities." <u>53-55% reduction in casualties</u>	Antonio - under dev & Ineq - hence need to imbibe Tribal Panchsheel principles in bringing peace & participative governance <u>Steps:</u> <u>Aspirational districts</u> <u>Road, Mobile tower</u> <u>PESA, FRA, Van Dhan-Jal Jangal Jameen</u> <u>Surrender & rehab project</u> <u>Joint Task force for interstate coordn</u>	SAMADHAN doctrine - Smart leadership Aggressive strategy - fortified police stations Motivation & training - Greyhounds Actionable intelligence Dashboard based KPIs Harnessing Tech Ex: UAVs, biometric smart guns A - Action plan for each theatre N - No access to finance Ex: PMLA
<u>Hybrid warfare</u>	Use of unconventional methods as part of multi domain operations without engaging in open hostilities Ex: Chinese apps harvesting sensitive		

	info, lone wolf, cyber warfare		
Role of external state & non state actors	State actors refer to foreign govts indulging in terrorism against another state to achieve own strategic & tactical objectives Ex: ISI, Syrian war Non state actors - are extended arms of the state that give it plausible deniability in operations Ex: JeM, LeT wrt ISI Through conventional & unconventional warfare, __ has been posing multifaceted security challenges for India		
Role of media & social networking sites	Social networking sites & media are increasingly being used to mould public opinions, spread disinformation and engage in illegal activities posing a threat to internal security	The Union Home Minister has given the strategy of “Trace, Target & Terminate” to end <u>organized crimes</u> “The new frontiers of war is the civil society which can be manipulated to hurt a nation’s interests.” - Ajit Doval	
Organized crime	Coordinated criminal activities by structured groups for financial gain.	New Bills replacing IPC, CrPC have separate provisions for OC - step in right direction	
Space Defence	Recently launched Mission DefSPACE - private participation	Build credible defensive & offensive capabilities + push for comprehensive & binding international agreement on peaceful use of Outer Space	US - dedicated Space force
NE Insurgency	Ongoing armed conflicts and movements seeking political and social change in India's northeastern states."	<u>Antonio quote</u> Peace in NE is key to safeguarding SUSI of India as well as achieving PM’s vision of “Act East, Act first through North East”	
Extremism	Advocacy of extreme ideologies or actions that challenge	<u>Antonio quote</u>	

	established norms and institutions usually through unconstitutional & violent methods		
<u>Terrorism</u>	Terrorism is the illegal use of violence or threats to spread fear among innocent civilians or governments for political or ideological goals. India victim for decades esp state sponsored	In light of increasing tech sophistication & globalization of terrorism, “War on Terror” demands <u>beyond-border</u> cooperation, suffocating <u>terror finance</u>, building resilient societies & robust institutions.	UK’s counter terror strategy of CONTEST - 4Ps - <u>Prevent</u> - people from being drawn - counter radicalisation <u>Pursue</u> - Stop a terror attack - detect, disrupt & prosecute Ex: NATGRID, NIA <u>Protect</u> - through access restriction - border security, cyber security Ex: CIBMS <u>Prepare</u> - resilience to counter an attack that cannot be stopped Ex: NSG Others - UN CCIT, FATF
<u>Linkages b/w OC & terrorism</u>	The convergence of OC & terrorism through resource sharing and mutual facilitation poses significant threats to our internal security and global stability.	<u>Template - war not of one agency or one govt alone.</u>	
<u>Linkages b/w development & extremism</u>	Poverty is the mother of crime - Marcus Aurelius. Thus underdevelopment and extremism are <u>inextricably linked.</u>	<u>"The battle against extremism begins with addressing the roots of underdevelopment and inequality." - Antonio Guterres</u>	
<u>Drug Trafficking & abuse</u>	India’s location between the golden triangle & crescent coupled with porous borders has made it highly vulnerable to menace of drug <u>trafficking & abuse.</u>	State is duty bound u/a 47 to prohibit the consumption of drugs due to safeguard social fabric, public health & internal security. Steps - Nasha Mukt Bharat - NIDAAN portal - NAPDDR - National Action Plan for drug demand reduction - NDPS Act & NCB	Operation Samudragupt - NCB & Navy seizure of Rs. 15000 cr in Arabian Sea Mansa Village PB - diverting youth energy into community work - “<i>Mera Pind Mera Maan</i>” 3 pronged reductions: OF - Demand - Supply - Harm

<u>Military reforms</u>	Recently, spate of reforms have been initiated to improve synergy, operational readiness & the combat edge of indian armed forces	Move towards comprehensive National Security Strategy encompassing Chanakya's 3 S - Shastra, Shaastra, Sanskriti. Implementation will be key.	
<u>Piracy</u>	Denotes criminal acts like robbery, hijacking, or hostage-taking targeting ships in international waters		
<u>Institutions - NIA</u>	NIA is the Central Counter-Terrorism Law Enforcement Agency of India	"The implementation is more important than their legislation" - Ajit Doval. Hence strengthen NIA wrt 4 F for protecting SUSI.	
<u>FATF</u>	Global watchdog that ensures intl cooperation & sets intl standards to tackle ML & TF.		
<u>CDS</u>	CDS is the highest-ranking military officer in India, responsible for coordinating and streamlining the overall military strategy and operations of the armed forces." > 70 countries have	The 21st CE security challenges of a 2 front war call for a proactive role of CDS in bringing synergy, jointness & integration + progressing towards a cohesive national security strategy.	Kargil Review committee + Shekatkar committee - Principal milit advisor, 4 star - Chairman - Chief of staff commtt - Head of dept of mil affairs - Member of DAC, Defence Planning Committee - Advisor to NCA (nuclear)
<u>Integrated theater command</u>	Synergising the capabilities & potential of the 3 services for greater integration & resource optimisation	Will be effectively only if supplemented with military modernisation and comprehensive national security strategy	Kargil Review committee recommended
<u>IPC changes</u>	Recently MoHA introduced 3 new bills to replace IPC, CrPC & IEA IPC - Bhartiya Nyay Sanhita Bill CrPC - Bhartiya Nagrik Suraksha Sanhita Bill IEA - Bhartiya Sakshya Bill	MoHA - the Spirit of the 3 new laws is not to punish but give justice to every citizen & prevent further crimes. Issues must be looked - - Hate speech removed - Detention time increased - Low bail focus - Vague terms - IPC equiv sectns	Rebributive, victim centric, Tech enabled, simplified IPC - new sections - S. 110 - org crime, - S. 111 - terrorism, - cyber crimes, mob lynching - 7 years - Police accountab CrPC - - Zero FIR - Mandatory videography

			- Timelines - chargesheet (90 days) & judgement (30 days) - Compulsory forensics - Community service IEA - - Admissibility of electronic / digital record
<u>Mob Lynching</u>	Refers to extra judicial killing of a person by mob action. <u>Ex: Udaipur, Una flogging</u> SC recently sought updates on action taken by govt against lynchings.	<u>Tehseen Poonawala case</u> - duty of state to protect citizens & “pluralistic social fabric” of country	<u>Steps:</u> - IT rules 2021, 2023 - Amendment to IPC, CrPC - RJ protection from lynching Bill 2019
<u>UAPA amendments</u>			
<u>Terror Finance</u>	Provision of funds/ financial support to individual terrorists or groups. Recently “No money for terror” conference held in New Delhi	Suffocating finance is necessary but not sufficient to tackle the hydra headed menace of terrorism. <u>Steps:</u> <u>UAPA Amendments</u> <u>Crypto KYC</u> <u>Dedicated cell in NIA</u> <u>Soft Demonetisation</u> <u>FCRA amended - NGOs</u> <u>FATF + PMLA</u>	
<u>Agnipath scheme</u>	Govt recently unveiled the Agnipath scheme to recruit youth into armed forces for short duration of 4 years.		

S&T

<u>TOPIC</u>	<u>INTRO CONCLUSION</u>	<u>Conclusion</u>	<u>VALUE ADDITION</u>
<u>General</u>		<u>RELEVANCE:</u> <u>Technologies -</u> <u>The benefits of __</u>	<u>Balasya Mulam Vigyanam</u> <u>Dangers: Technology cannot be</u>

		tech are especially meaningful for India at a time when _____. (IR 4.0) With a solid research base & workforce founded on reliable govt support, India can harness innovative applications - stimulating economic growth & job creation which will feed back into the tech economy. <u>RESPONSIBILITY:</u> As we continue to harness the power of _____, we must do so responsibly, considering ethical, safety, and environmental implications.	uninvented "The future depends on what we do in the present." - Mahatma Gandhi Lab to field through industry academia collab + incentivise adoption - so that it benefits not only us but generations to come + facilitate PPP in R&D Regulatory sandbox Issues: - Skill, Money, IPR - Security, Privacy, Divide - Ethics, Employment, Env't 70% jobs threatened by automation in next 20 years "Jai Jawan, Jai Vigyan, Jai Anushandhan" - PM Modi
Space Sector	Intro: Recently Govt released the Indian Space Policy 2023	Vision of India as the launchpad of the world	95% Global space economy opps are in ground based operations & space based services - neglected in India due to too much focus on launch & manf (5%)
Private players in Space	Intro: India led by ISRO occupies a meager 2% share (5 Billion \$) in the global space economy of 450 Billion \$ - indicating the huge potential & need for private participation. Ex: Agnikul	Indian space economy will become 60 Billion \$ by 2030 from present 5 Billion \$ - presenting a golden opportunity Draft Space Activities Bill 2017 INSPACE (interface with pvt & ISRO & regulator), NSIL (aggregator)	ISRO Budget - only 1.5 Billion \$ Abhi 30K satellites - projected to be 80K by 2032 ISRO - 50K \$ / kg vs SpaceX charges 3K\$/kg Opened since 2000 for Pvt
Indian Space Policy 2023	Recently released	60 Bn \$ space economy of India from 5 Bn \$ presently	<u>Pvt participation - NSIL</u> <u>Regulatory body - inSPACE</u> <u>Nodal Dept - DOS</u> - policy guidelines <u>Rationalization of ISRO's role</u> <u>End to end Pvt activities like - ground stations, launch vehicles etc</u>

<u>Space debris</u>	Refer to Defunct human-made objects and fragments that orbit the Earth	NETRA for situational space awareness by ISRO	100 million pieces of > 1mm in space 2009 accident - US & Russian satellite Indian ASAT test - produced 400 pieces of space debris UK - RemoveDEBRIS mission Lower orbit - slow down, higher - graveyard orbit
<u>Aditya L1</u>	Space based solar observatory that will be launched by India & place at Lagrange point L1	The __ mission is not just about unraveling cosmic mysteries but also pivotal to India's vision of space leadership.	Payloads - SUIT - Solar ultraviolet Imaging telescope, ASPEX Applications: Nuclear energy, GPS & satellite radio communication (solar storms)
<u>Chandrayaan</u>	Successful 2nd attempt at soft landing on south pole India 4th nation after US, Russia & China;	PM - Win for humanity & science - ARTEMIS mission provides opportunity <u>Why the south pole?</u> 1. Unexplored regions 2. permanently shadowed - possible to find water 3. cryo preserved evidences	Payloads: - ChaSTE (thermal mapping lunar regolith), - RAMBHA (near surface plasma density), - ILSA: Lunar Seismic Activity, etc Components: - Propulsion module + Vikram + Pragyan - Objectives: 3 fold : Demonstrate soft landing + rover's loitering capabilities + In-situ observation Improvements : - “failure based design” from “success based design”: - Landing area widened Legs Strengthened, Fuel Increased for last minute changes, Additional instruments
<u>Gaganyaan</u>	ISRO confirmed unmanned mission will take place next year making India 4th nation	Same as Aditya L1	Features: - circle Earth for 5-7 days at LEO about 300-400 km from earth - 3 flights will be sent into orbit - 2 unmanned & 1 human spaceflight (3 astronauts) - crew + habitable orbital module + Life support + Human rated GSLV Mk 3 + re-entry & recovery system + escape abort system
<u>Cryogenics</u>	Production &	Refer template	Applications:

	behavior of materials at very low temperatures - below -153 degrees		- MRI - Freeze drying - Cryosurgery - Cold treatment for tools - metallurgy - Cryo engines (liquid H2, O2)
Semiconductor	Materials which have conductivity between conductors & insulators - & form heart & brain of all modern electronics	Steps taken like India Semicon Mission, PLI & DLI. Natl Implementation will be the key.	India imports all chips Abhi 25 billion \$ - will touch 100 Billion \$ by 2025 Issues - highly concentrated global supply chain (47% with US) Suggestions: - BEL & HAL as semicon fab foundry - Focus on ATP - assembly, testing & packaging - low cost technical manpower - Leverage QUAD semiconductor supply chain initiative for tech transfer
Cyber physical systems	System in which mechatronic systems are controlled or monitored by computer based algorithms.	National mission on interdisciplinary cyber physical systems will help in augmenting human capabilities (prosthetics) & bringing large scale efficiency Ex: automated logistics	Ex: autonomous cars - collision avoidance, robotic surgery - precision smart grids Features - Miniaturization, communication n/w (devices talking), Computation ability
IOT			
5G/6G	Fifth/ Sixth generation wireless communication technology designed to connect everyone & everything	Success depends on govt's ability to bridge digital divide & viability of telecom companies to invest. Steps: National Digital communication technology 2018 NOFN	Cost of data - 300 rs to 10 rs per GB Millimeter wave spectrum - allows more connection density Issues: - Low fibre backhaul - 80% towers connected through microwave backhaul - IPR blocking tech access - Hardware - changing phones + imported - Security Applications: - Smart cities, eGov - automated factories (ferrari), - Seamless coordination Ex: NATGRID - Entertainment & multimedia

			industry - Ex: Immersive - <u>multiverse</u>
<u>Quantum computing</u>	Application of collective properties of quantum states like superimposition & entanglement to achieve various objectives far beyond traditional capacity Ex: Qubits	Quantum research facility setup at Mhow, <u>QUEST - Quantum enabled S&T prog 2018</u>	Ex: Quantum supremacy : Google Sycamore - 200 seconds vs supercomputers 10k years <u>Advantages: SSSP</u> - <u>Speed</u> , - <u>storage</u> , - <u>simulation</u> , - <u>privacy</u> <u>Applications:</u> - <u>Research</u> Ex: Genome India project - <u>Secure Communication</u> - <u>MICIUS CHINA</u> - <u>Pharma</u> - predict epidemics - <u>Meteorology</u> - weather forecasting, climate models - <u>Data processing for industry</u> - ex: consumer behavior, financial markets
<u>Blockchain</u>	Decentralized immutable digital ledger distributed across a network of computer systems. Transactions are anonymous & secure without the need of 3rd party. Encrypted Ex: Crypto currency	<u>Steps:</u> <u>IndiaChain</u> by NITI Aayog for public records National strategy on Blockchain 2021 by MeITY <u>Dubai experience</u>	<u>Applications:</u> 1. <u>SCM</u> - Ex: IBM Food Trust Blockchain - tracks source of food products contamination 2. <u>Medical records</u> Ex: AYUSHMAN Bharat Digital Mission 3. <u>Property records</u> Ex: NRLMP 4. <u>Asset protection</u> Ex: NFT - indisputable record of real time ownership - IPRs 5. <u>Good governance</u> Ex: Digital certificates 6. <u>Corruption</u> Ex: procurement, DBT 7. <u>Voting</u>
<u>AI</u>	<u>AI</u> - Simulation of human intelligence by machines like capacity to learn & make decisions Ex: pattern recognition, predictive analysis <u>Machine Learning</u> - Subset of AI - that uses statistical learning algos Ex: Netflix, Spotify	AI is poised to add 500 Bn \$ to India's GDP by 2025 (10% of 5 Trillion \$ target) Regulation must strike a balance between encouraging innovation & preventing unintended consequences. (privacy, digital divide) ASILOMAR	3 cognitive skills: 1. Learning 2. Reasoning - 3. Self correcting <u>Applications:</u> 1. <u>Health</u> - Diagnosis Ex: 3Nethra eye scanner by Microsoft 2. <u>Agri</u> - Autonomous farming, Predicting pest attacks, KN APMC - price forecasting 3. <u>Education</u> - Adaptive learning Ex: Robokart 4. <u>Smart cities, Mobility, Grids</u> 5. <u>Marketing</u> Ex: personalized

	<u>Deep Learning</u> - ML technique that is inspired by how a human brain filters info. Relies on Artificial Neural Networks	principles for AI - guiding light for AI development in India and world <u>AI for ALL principle</u>	recommendations, Alexa 6. <u>SCM</u>
<u>Generative AI</u>	Type of AI that can produce realistic content like text, images, audio, based on data they've been trained on Powered by GAN - Generative Adversarial Networks & enabled by Transformers, LLMs, deep learning Ex: CHATGPT, Dall-E	Tech is a double edged sword - Alan Moore. Clerical jobs at greater risk. Need to evolve capabilities but along with that discussion needed at international high table to confront the new realities.	<u>Applications:</u> - ChatBOTS - DeepFAKES - Dubbing - Product demonstration - Designing <u>India Ineq report by CMIE-</u> Internet penetration - urban 67% vs rural 31%
<u>Web 3.0 Metaverse</u>	<u>Web 3.0</u> - New iteration of WWW - <u>Metaverse</u> - 360 degree Immersive experiences brought by integrating physical, augmented & virtual reality in a shared online space	-	-
<u>Crypto currency</u>	A digital or virtual form of money that uses cryptography for secure transactions and operates independently of traditional financial institutions <u>Ex: Bitcoin</u>	Regulation must balance.... CBDC is in right direction <u>Diagram - Complexities + Potential</u>	<u>Features: ABCDEF</u> - Anonymous - Blockchain - Cross border flow - Decentralised - Encrypted - Fast RBI - next global crisis from crypto Suggestions: reg sandbox, stablecoin, FSB intl collab
<u>NFT</u>	Unique digital	<u>Same as crypto</u>	-

	assets that represent ownership of a specific item or piece of content, using blockchain technology to verify their authenticity		
<u>CBDC</u>	Legal tender issued by Central bank in digital using blockchain platform	Need for wider stakeholder participation in rigorous analysis of outcomes of recent pilot projects conducted by RBI to ensure full operationalisation in near future	- Financial inclusion - Smart contracts
Biotech	Biotech refers to the use of biological systems & living organisms to develop various products India administered 220 cr vaccines + eliminated smallpox, polio BP: COVID Suraksha Program success - World's first covid nasal vaccine - BB (success of state led R&D)	Despite tremendous growth, India's share only 3% of the global biotech industry - huge potential <u>Steps:</u> - BIRAC - National Biotech dev strategy 2015-2020 - National Biopharma mission	Need to diversity (Biopharma - > 50% share) <u>Applications:</u> - <u>Agri</u> - GM, Sex sorted semen - <u>Nutrition</u> - bio-fortification - <u>Env't</u> - bioremediation Ex: Biodigestor - <u>Industry</u> - biodegradable plastics, biofuel, fermentation, enzymes - <u>Health</u> - vaccines, molecular diagnosis like ELISA, stem cell therapy, recombinant DNA tech <u>Issue:</u> 1. Biological weapons/ Bio-terrorism 2. Biopiracy 3. Biosafety 4. Patents
<u>GM Crops</u>	Recently Govt approved envtl release of GM Mustard Hybrid DMH 11 Technique of changing characteristic of an organism by transferring a piece of DNA from another organism 5th largest area under GM	<u>Precautionary principle</u> Instead of GM --> "safe agri ecological solutions" Ex: system of mustard intensification	DMH 11 - 20-25% higher yield (presently yield ½ of global average) BT Cotton helped reduce insecticide use by > 50% in AP & GJ

	<u>cultivation</u>		
<u>Nano tech</u>	Field of research & innovation concerned with building, designing & modifying things at scale of atoms, molecules at nanometer range (1-100 nanometer where 1 nanometer = 1 billionth of a meter)	<u>Lab to field</u> <u>Steps:</u> Nano S&T Initiative (<u>NSTI</u>)- umbrella - infra, intl collab, cap building <u>Nano Mission 2007</u> <u>Visheswaraya PhD scholarships</u>	<u>Applications:</u> 1. <u>Agri</u> Ex: nanosensors, nano urea, precision farming, nanogels for pest control, sustainable agri 2. <u>Health</u> - a. <u>Controlled Drug delivery</u> b. <u>Remote patient monitoring</u> c. <u>Nanobots - artery blockage</u> 3. <u>Space</u> 4. <u>Energy</u> - solar paints, nanogenerators - flexible thin sheets 5. <u>Construction</u> - strength, water seepage resistant, heat resistant molecules 6. <u>Electronics</u> - Nano materials/batteries - improved storage capacity, speed, weight 7. <u>Defense</u> - stealth
<u>IPR</u>	legal protections for original creations, inventions, and ideas	Reform IPR ecosystem to balance incentives for cutting edge research with larger public good	
<u>Agritech</u>	MMS - agri suffers from technological fatigue	Not just growing more but growing smarter & more sustainably	<u>Applications of each tech as discussed above</u>
<u>EVs, Lithium Ion batteries</u>			
<u>IR 4.0</u>	ongoing era of technological advancement characterized by the fusion of digital, biological, and physical technologies Ex: disruptive dozen	<u>Balasya Mulam Vigyanam</u> In the fourth industrial revolution, it's not about the big eating the small, it's about the fast eating the slow	
<u>Personalities</u>	____ was a great scientific luminary, whose profound contributions in the field of ____ have reshaped our understanding and spurred unprecedented	____'s contributions have left an indelible mark in the field of ____ and continue to influence countless advancements & innovations that shape our modern world	

	<u>advancements.</u>		
<u>IT</u>	Use of computers, software, and networks to handle and share information efficiently.	<u>IT + IT = IT</u> <u>Common for IT related:</u> The vision of a developed & digitally empowered India by 2047 calls for urgent investments in _____	
<u>Drones</u>	Aircrafts without human pilot - also known as unmanned aerial vehicles Recently Drone (Amendment) Rules 2022 notified - abolishing reqt. of drone pilot license	<u>Steps like:</u> <u>Digital Sky</u> platform, <u>PLI</u> scheme, <u>Mission Drone Shakti</u> for (DrAAS) Drone as a service in Budget 2022-23 <u>Anti Drone tech</u> by DRDO Drone rules - Drone classification, air zone classification, drone corridors	<u>Applications:</u> 1. <u>Healthcare</u> - TL 'Medicine from sky' initiative 2. <u>KISAN DRONES</u> , <u>SWAMITVA</u> 3. <u>Law & order</u> Ex: Delhi Govt using to record communal violence 4. <u>Delivery</u> - Amazon's Prime Air 5. <u>DM</u> - Used in 2022 AS floods 6. <u>Security</u> - Swarm drones, loitering munition 7. <u>Law enforcement</u> Ex: illegal mining
<u>National Green Hydrogen Mission</u>	Producing H2 from water using only green & renewable sources. Black or brown-coal <u>Grey - natural gas</u> <u>Blue- carbon capture</u> Centre cleared natl. green hydrogen mission of 19K cr:	Any RE question - Game changer in reshaping our energy landscape, mitigating climate change while fueling economic progress	<u>Goals by 2030:</u> A) <u>5 MMT</u> of green hydro. B) <u>SIGHT</u> (strategic interventions for green hydrogen transition prog) - financial incentives for manufacturing electrolyzers. <u>Advantages:</u> 1) reduce fossil fuel imports - Rs. 1 lakh cr 2) decarbonise - industrial, mobility, energy sectors. 3) global hub for derivatives <u>Issues:</u> - India - <1% of global prod - immature tech - 3 times more expensive than Black H2 - CEEW
<u>Women in STEM</u>	<u>Ritu Karidhal - Chandrayaan</u>	PM Modi "our thinking is not merely to empower women through science but also empower science by contribution of women"	

Disaster management

Topic	Introduction	Conclusion	Data
<u>Templates</u>	<u>Impact</u> - socio economic - infra - env - Epidemics and pandemics - psychological - political / governance	Disasters are inevitable, but their impact is not. Through robust disaster management strategies, we can mitigate risks, save lives, and build more resilient communities	<u>Indian Framework</u> - DM act : 3 tier structure - NDMA ; SDMA ; DDMA - NDMP 2016 - aligns India with Sendai - NDRF - National disaster response fund - national calamity contingent duty <u>SENDAI</u> - reduce 4 : 1) mortality 2) affected ppl 3) economic loss 4) damage to critical infra and basic services - increase 3 : 1) countries with DRR strategies 2) int coop 3) access to multihazard EWS
<u>Keywords</u>	DRR - systemic development and application of policies and practices to minimise vulnerabilities of the society in the broad context of sustainable development	- Vulnerability mapping - Prevention and mitigation - Emergency response - Loss and damage - Local genius - First responders (OD balasore train accident)	- 3R's : relief ; rehabilitation ; reconstruction - learning curve - disaster package politics - BBB
<u>Disaster</u>	A disaster is defined as disruption on a massive scale , either natural or man made , occurring in a short or long period. It affects the capacity of the affected society to cope with using its own resources		- India is vulnerable to 30 different types of disasters - EQ - 60% - cyclones -12% - floods - 12% - drought - 68% - 5700/7500 km - cyclone and tsunami
<u>Earthquakes</u>	Shaking of the earth's surface caused by seismic waves generated due to sudden release of energy in the earth's crust or upper mantle	WF : EIA before mining and dam constriction Bhungas - kutch Ikra bamboo houses - SK	- Major earthquakes - GJ 2001 ; nepal 2015 ; SK 2011 - 60% of india's land vulnerable to earthquakes (80% popln) - steps taken

			1 national retrofit program 2014 2. National EQ risk mitigation project 3. EQ disaster risk indexing Best practice - Japan : 80% compliance of quake resistance building Statute of unity
<u>Earthquake swans</u>	Series of low magnitude earthquakes that occur in a localised region over a period of time raging from days to months No clear sequence of foreshocks , main and after shock Difficult to predict		<u>Palghar MH ; 30 low intensity EQs in 2 months</u> <u>Causes</u> - hydro siesmicity - magnetic activity - weak crust
<u>Heatwaves</u>	Period of abnormally high temperature over certain areas which could be fatal to humans and animals	<u>WF : NDMA guideline</u> - sensitive approach : identifying most vulnerable Nagpur - heat action plan + heat wave index Ahmedabad - cool roofs prog for slums + colour coded heat alerts	<u>IMD : >40° plains ; >27° coastal ; >30° hills + 5-6° above normal</u> 2015 new delhi heat wave 1800 died 2022 - europe , north america ; china - >20 k people died WB : 75% Indian LF - heat exposed labour + heat wave situation in india could break human survivability limit
<u>Cold wave</u>	Rapid fall in temp within 24hrs to a level requiring substantially higher protection to agri industries and social activities	<u>Rein Bashera in delhi</u> Prepare an action plan	<u>Conditions</u> - plains : <10° + 4.5° below normal for 2 days consecutively IMD - 3 times increase in no o coldwave days -2017-20
<u>BOMB CYCLONE ; BLIZZARD ; SNOW STORY</u>	Rapidly strengthening storm where pressure drops 24mb in 24 hrs		
<u>Landslides</u>	Downward and outward movement of slope material like rock , debris or earth down a slope under influence of gravity	<u>WF :</u> - NDMA - NLRMS (risk mitigation strategy) - national landslide risk mitigation project 1) GSI : landslide	2021 - kinnaur HP 2013 UK flash floods 2018 KL landslide <u>Vulnerability</u> - GSI - 12% land area prone - 2 areas prone : himalayas

		hazard zonation maps using UAV 2) microzonation 3) capacity building	- NE + WG - 70 HEP in himalayan states : EQ triggered landslide
		MH - geo textile Uttarkashi - EWS + stablisation	182 landslides in 10 states in 2022
VOLCANO ES	Volcano is an opening in the earth's crust from which molten rock (magma) , gases and debris erupt to the surface		+ve effects -soil fertility - geo thermal energy ; hot springs - Minerals - cooling effect
TSUNAMI	Series of extremely long waves caused by a large and sudden displacement of ocean water due to EQ ; volcanoes etc	WF DOARS : deep ocean assessment and reporting system INCOIS : by MoES setup indian tsunami early warning centre (ITWEC)	2004 tsunami - 2.3 lakh people died
		Sagar vani app	
CYCLONE S	Rapid inward air circulation around a low pressure area . cyclones have become a perennial phenomenon in India	IMD - colour coded warning	Conditions -temp >27° ; corolios force ; small variation in vertical wind speed ; low pressure ; upper air divergence
	Silent killer	OD model	
		RIMES (regional integrated multihazard EWS)	Why increases in arabian sea
		Multiple engagement areas	-heating : IOD ; GHG
		-natural storm brakes (mangroves)	- El nino
		- tech : UAV ; satellite	- weakening of monsoon
		- coordination	- Low wind sheer coz of increase in CO2
			5700/7500 km prone to cyclone
			Data
			-10% of world's tropical cyclones
			-13 states and UTs affected
			- costs : 2% of GDP
			- cyclone Nirvar - damaged 23000 acres of crop
STORM SURGE	Abnormal rise of sea level as the cyclone crosses the coast leading to inundation of the coastal strip		
TEMPERAT E CYCLONE	Cyclones which develop in the middle and high latitudes (35° -65°) N&S		Polar front theory

S	beyond the tropics		
<u>DROUGHT</u>	Period of abnormally dry weather sufficiently prolonged for lack of water causing serious hydrological imbalances in the area	-Not defined as disaster under DM act - Drought crisis management plan , 2015 1)MGNREGA to provide employment to affected 2) action to recharge ground water 3) crop loss compensation	-70% draught prone - 20% india's land area facing draught like conditions 4 types - Meteorological - Hydrological - Agricultural - Ecological Schemes -PMGSY - soil health card - PM fasal bima
<u>FLASH DROUGHT</u>	A flash drought is a term used to describe a rapid and intense onset of drought conditions over a relatively short period of time, usually a few weeks or months. Unlike traditional droughts that develop more gradually over longer periods, flash droughts can emerge suddenly due to a combination of factors such as high temperatures, reduced precipitation, low soil moisture levels, and increased evaporation rates.	Seechwal model Israel case study Agro climatic agriculture	Study - India could experience more flash droughts by the end of the century
<u>FOREST FIRES</u>	Major fires occurring unpredictably in combustible environments like dry forest and burns for long term over a large area	-National action plan on FF 2018 - FF alert system - i-Forest initiative	Vulnerability -FSI : 36% forest vulnerable Examples : kullu valley FF 2021 FSI : FF cost : \$1.5 bn
<u>FLOODS</u>	Situation of high water level along a river or on the coastal areas that leads to inundation of land	NDMA guidelines -structural measures : reservoirs ; desiltation ; afforestation - non structural : flood plain zoning ; EWS BP : -buoyant buildings netherlands - bamboo house NE - Rain gardens ; sponge city	-National flood commission : 40 mn ha prone to floods ; cost \$7 bn -80 % rain in 3 months + cloud burst -12% India vulnerable - India : 1/5th of global flood deaths

		- jabalpur : rebate on property tax for RWH	
<u>URBAN FLOODS</u>	Inundation of infrastructure in densely populated urban areas due to intense rainfall which overwhelms the capacity of drainage systems	<u>IFLOWS - mumbai RWH</u>	Germany : Sponge cities : Hamburg
<u>CLOUD BURST</u>	Short duration intense (100 mm/hr) over a small area	BP : copenhagen cloudburst formula - blue green infra ; city planning	UK 2021 JK 2010 Vulnerable terrains - mountains ; deserts ; interiors of landmass Hotspot : CNP - world's most vulnerable spot for formation of severe thunderstorm
<u>LIGHTNING STRIKES</u>		LRIC (lightening resilient india campaign) - IMD , NDMA and others : aim : reduce no of deaths to less than (1200 / annum)	1.85 cr LS recorded b/w 2020-21 "Damini " IITM pune - 3hrs advanced warning
<u>GLOF</u>	Breach of glacial moraine wall causing flooding due to increased water content higher than the capacity		2013 - kedarnath tragedy 2021 - UK Chamoli Vulnerability - 3rd pole - 8800 glacial lakes - CC - Tectonically active
<u>NUCLEAR DISASTERS</u>		<u>Public liability insurance act</u>	Vulnerability -22 functioning nuclear reactors - shares boundary with 2 nuclear states - nuclear reactors on coast line ; kalpakkam NPP flooded by tsunami
<u>EXTREME WEATHER EVENTS</u>	IPCC : those events that are rare at a particular place and time		CSE report 2022 -EWEs recorded on 241/273 days - india recorded warmest , driest and wettest days in 2022 - 75% districts affected with EWEs
<u>Innovative suggestions</u>	-Safety culture - Decentralised , pro active ; preventive , community resilience , self sufficiency ; parallel rather than equential ; equitable	-Every \$ spent on mitigation saves \$7 on response and recovery - sendai - 0 casualty approach - funding : catastrophe bonds	-resilient Infra - SDG 9 ; CDRI - EWS and info dissemination : OD SATARK mobile app

<u>Conclusion</u>	Tehri Dam case : SC - <u>DM is part of right to sustainable development</u>	The culture of “ prepare and prevent l not repair and repent “ must become universally engrained for a prosperous and resilient India	Risk informed development and holistic disaster governance
<u>WF</u>	Right wing wants less vat on LSD drug <u>RW wants LES VAT LSD</u>	<u>R - risk coverage</u> <u>W - women role</u> <u>L - leadership</u> <u>E- EWS</u> <u>S - sendai</u> <u>V- vulnerability mapping</u> <u>A - academia</u> <u>T - tech</u> <u>L - local capacity</u> <u>S - social media</u> <u>D - DRR</u>	